

NON-CF BASED VALUATION METHODS

(SELECTING PEERS, TYPES OF MULTIPLES, ASSET-BASED VALUATION APPROACHES)

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Source:

- Juhász ppt + videók
- Valuation_Measuring_and_Managing_the_Value_of_Comp..._---_(18._Using_Multiples)

Alternatively to, or instead of DCF methods we can use non-DCF methods, which primarily include multiple-based valuation and asset-based valuation.

Multiples

There are two multiple-based methods:

- Comparable transactions
- Market multiples

Comparable transactions:

- look at *historical M&A transactions* to figure out at what price similar companies (sector, size, geographic location etc.) were sold/bought
- There may be *distortions* because a transaction done in 5 years probably has nothing to do with the current situation (market changed over time). Moreover, the purchase price of the firm could also contain some synergy (thus not reflecting purely its value) premium, thus the price not reflecting the pure value of a company.

Market multiples:

- look at *current pricing of the peers* on the stock exchange to estimate multiples
- There may be *distortions* when checking current multiples of listed peers bc *market efficiency* (prices reflecting all available information and real values) is questionable, potentially leading to mispricing in valuation. Moreover, current stock prices reflect the price for a small number of stock trades which entail *minority discount*.

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Advantage: both are quick, simple, and reflect the current market opinions/assumptions.

Disadvantage: implicit assumptions, quality of data questionable (especially in case of comparable transactions), comparability (it requires more knowledge and understanding regarding the results), bubbles, any value could be justified.

Valuation process

The valuation starts with **choosing the multiple**. When choosing from several multiples, choose *simple* (easy-to-understand), clear and accessible (data available) ones. You also have to figure out which multiples describe best the population of an industry (*relevance*). Have a *consistent* ratio (same dimension for numerator and denominator: either whole firm, either equity). Have the multiples defined the same way for all peers (*uniformity*) – same year, currency, method etc.

The challenge is **choosing the right peers**. We choose our peers based on the selected multiples! We choose comparable firms which are similar in risk, growth, profit generation etc. A good peer group consists of companies that not only *operate in the same industry* but also have *similar prospects for ROIC and growth*. The common practice is to select a group of 8-15 peers and take the average of their multiples.

However, we have to **control for differences** (subjective correction, modified multiples), bc the average value of peers may still be different from our value. We can *recalculate the multiples* based on our characteristics or use *discount/premiums* on the original results (when peers multiples were used).

Finally, we have to **check whether the results are realistic**.

Mistakes:

- Description power of multiples is not checked → choose ratios that are applied in your industry
- Calculation of multiples is not standardized (calculation of multiples was not done the same way for all the firms) → download the raw data and calculate the multiples yourself
- Peers are competitors rather than similar firms (even though they are competitors, it does not mean they are totally similar)
- Necessary discounts not applied → the peers' average is usually different, so when using their multiples, control for differences

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Grouping of multiples:

- Profit-based
- Revenue-based
- Asset-based
- Industry specific

Profit-based multiples:

- The value of the equity / value of the whole firm is contrasted to some kind of profit. There are 3 requirements: similar CF-generation, and similar risk of generating this CF, similar stable long-term growth rate (g). *These 3 assumptions apply to all the groups.*
- ex. P/E, EV/EBIT

Revenue-based multiples:

- Revenue is rarely negative, so we do not have to drop potential firms, like in case of profit-based multiples → ends up w/ a better group because if bad companies are dropped (negative profit), then our firm would be contrasted only to profitable (best) companies across the industry.
- One additional assumption: firms generate a similar amount of CF from one unit of sales (similar margin).
- They are easy to use, and it is easier to get data on sales. The accounting limitations are low (even with different accounting regulations the revenues are barely distorted).
- ex. EV/S

Asset-based multiples:

- The BV of equity / BV of IC contrasted to the MV of equity of MV of whole firm.
- The fourth assumption here is that capital efficiency ($ROI(c)/ROE$) should be similar to peers.
- They are stable in the long run and easy-to-compare, but accounting policy differences (ex. depreciation method) could lead to distortion in BV of equity and BV of IC. So, avoid contrasting firms with very different accounting backgrounds.
- ex. P/BV, Tobin's $Q = MV \text{ of Assets} / \text{Replacement value of Assets}$. The problem is that neither of these can be observed on the stock market.

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Industry-specific multiples:

- With these multiples we value the whole firm, not its equity. We have to find some *physical quantity* (usually not in financial statements) that has a very strong link to the value of the firm. The requirement is that all the peers are in the *same industry*, they use the *same technology*, and they have the *same efficiency* as our company.
- Regarding these ratios no accounting issues may emerge. There is no need for extra further information.
- ex. telecommunication sector: number of subscribers; number of clicks for internet.

Terminal value for non-CF based valuation methods

- **Liquidation method:** after the explicit period we liquidate the firm. This is the worst case scenario - we will sell all the assets. If we assume that in the terminal value period no projects will generate positive NPV (bc $ROIC = WACC$), not even the old ones will generate CF anymore, then the value of the firm will be equal to IC, which is the value of the assets owned.
- **Replacement method:** we see that the company will perform well in 10 years' time, but we don't really know how well it will, so check what the assets that will be owned in 10 years' time will be and surely the company will be worth at least that amount.
- **Multiple-based method:** hardest to defend from theoretical point of view. Hard to judge how our and other similar firms would look in 10 years' time and what the multiples would be. People look at how the company will look in 10 years' time and check what kind of multiples the company would have today if it looked like that. This would assume that market conditions would not change over the next 10 years.

Asset-based valuation

This approach *focuses on the assets that the company has*, so we do not value the business. But the method is still important because it can *highlight whether it is better to stop the operation* and liquidate the firm or continue the operation. The value of the company in this case contains the

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value of operation (current main activity) + *value of additional assets* not needed for the current activity:

$$V = V_{OP} + V_{Other}$$

The challenge is to **identify what assets are**.

In ACCOUNTING:

- *Owned by the firms*: 3 rights (to use it, to modify it, to sell it) if an asset is used
- Expected income exceed expected costs
- These could be predicted with acceptable uncertainty (absolutely guaranteed)

In FINANCE:

- It generates future CF
- If not, it protects from generating negative CF
- Option-like assets, having some chance to get in the 1st or 2nd category
- Quasi assets: assets only manageable, but not owned (cannot sell them) (ex. good connection with suppliers)

Asset valuation methods

Type of value	Time orientation of data used		
	Past	Present	Future
Entry	Purchase cost	Substitution value (repurchase, recreation) Replacement value	–
Exit	–	Net realisable value, Liquidation value	DCF, Capitalisation, Real options

Past: modified purchasing cost (book value in accounting)

Present: how much would it cost to buy it today

- Substitution value: getting the same functionality as the asset is able to provide now
- Replacement value: buy the same asset; we assume the company will continue to operate similarly and requires estimating the cost of acquiring the same assets

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Present: for how much it could leave the firm

- Net realizable value: the price at which we could sell it (minus transaction costs, tax)
- Liquidation value: we assume that we have to sell the asset rapidly (higher transaction costs or lower sell price); we assume that the ongoing operation are in danger

Future

- DCF: predict CFs of the asset and discount them back
- Capitalization: a type of multiple-based valuation for assets (typically for office buildings)

Accounting value	Business value
Past oriented	Present and future focused
Based on documents	Predictions, estimations
Rule of prudence	Expected value
Aims to inform outside stakeholders	Made for owners and managers

Difference between the 3 major valuation approaches:

- DCF focuses on getting the value of a predicted future operation
- Multiples would estimate the price we might receive if a typical average buyer from the market would acquire the firm
- ABV estimates how much money we might be able to withdraw if we stop the operation of the firm, liquidate the assets and pay back all the liabilities